EXPERIMENT 5

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Experiment 5 – SQL Conditional Logic (Odd & Even Values)

SQL Conditional Logic using the MOD (%) operator to classify employee salary values as Odd or Even. This experiment demonstrates numerical analysis and conditional evaluation using SQL queries in Oracle and PostgreSQL.

Aim

To understand and apply conditional logic in SQL by using the modulus operator (MOD / %) to analyze numerical data and classify employee salaries as odd or even, thereby improving data analysis and decision-making skills in SQL.

Objective

- To write and execute SQL queries using the MOD (%) operator.
- To retrieve salary details from an employee table.
- To classify salaries into Odd and Even categories.
- To display odd and even salary values separately.

Software Requirements

Database Management System:
Oracle Database Express Edition (Oracle XE)
PostgreSQL Database

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Database Administration Tool / Client Tool:
Oracle SQL Developer (for Oracle XE)
pgAdmin (for PostgreSQL)

Practical / Experiment Steps

1. Create an employee table with columns: id name department salary
2. Insert sample employee salary records into the table.
3. Write a SQL query to retrieve employee salary details.
4. Use the MOD(salary, 2) function (Oracle) or salary % 2 (PostgreSQL) to check whether the salary is odd or even.
5. Use a CASE statement to classify salaries as:
   'Odd Salary'
   'Even Salary'
6. Execute separate queries to:
   Display only employees with Odd salaries.
   Display only employees with Even salaries.
7. Observe and record the output results.

Procedure

1. Open Oracle SQL Developer or pgAdmin.
2. Connect to the database.
3. Create the employee table using CREATE TABLE.
4. Insert sample records using INSERT INTO.
5. Execute SELECT queries with the MOD operator.
6. Apply CASE statement for conditional classification.
7. Display results and capture screenshots.

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Input / Output Details

Input

Table Name: employee
Columns: id (Integer)
name (Varchar)
department
(Varchar) salary
(Integer)

Logic Used:
MOD(salary, 2) or salary % 2 CASE statement

Step-wise Output

Step 1 – Create employee Table

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Query Query History

```sql
CREATE TABLE employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(100),
    salary INT
);
```

Data Output Messages Notifications

```
CREATE TABLE

Query returned successfully in 201 msec.
```

Step 2 – Insert Data into employee Table

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No limit

Query Query History

```sql
INSERT INTO EMPLOYEE VALUES (101, 'Amit', 25000);
INSERT INTO EMPLOYEE VALUES (102, 'Riya', 32001);
INSERT INTO EMPLOYEE VALUES (103, 'Rahul', 45000);
INSERT INTO EMPLOYEE VALUES (104, 'Sneha', 28999);
INSERT INTO EMPLOYEE VALUES (105, 'Arjun', 50000);
```

Data Output Messages Notifications

```
INSERT 0 1

Query returned successfully in 248 msec.
```

Total rows: Query complete 00:00:00.248

Step 3 – Display Employee Salary Records

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Query Query History

```
SELECT * FROM EMPLOYEE;
```

Data Output Messages Notifications

Showing rows: 1 to 5

	emp_id [PK] integer	emp_name character varying (100)	salary integer
1	101	Amit	25000
2	102	Riya	32001
3	103	Rahul	45000
4	104	Sneha	28999
5	105	Arjun	50000

Successfully run. Total

Total rows: 5 Query complete 00:00:00.473

Step 4 – Classify Salaries as Odd or Even

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Query Queries History

```sql
SELECT EMP_ID,
       EMP_NAME,
       SALARY,
       CASE
           WHEN SALARY % 2 = 0 THEN 'Even Salary'
           ELSE 'Odd Salary'
       END AS SALARY_TYPE
FROM EMPLOYEE;
```

Data Output Messages Notifications

Showing rows: 1 to 5

	emp_id [PK] integer	emp_name character varying (100)	salary integer	salary_type text
1	101	Amit	25000	Even Salary
2	102	Riya	32001	Odd Salary
3	103	Rahul	45000	Even Salary
4	104	Sneha	28999	Odd Salary
5	105	Arjun	50000	Even Salary

Total rows: 5 Query complete 00:00:00.156

Step 5 – Display Only Odd Salaries

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Query Query History

```
SELECT * FROM EMPLOYEE
WHERE SALARY % 2 <> 0;
```

Data Output Messages Notifications

Showing rows: 1 to 2

	emp_id [PK] integer	emp_name character varying (100)	salary integer
1	102	Riya	32001
2	104	Sneha	28999

Successfully run. Total qu

Total rows: 2 Query complete 00:00:00.181

Step 6 – Display Only Even Salaries

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```
SELECT * FROM EMPLOYEE
WHERE SALARY % 2 = 0;
```

Data Output Messages Notifications

Showing rows: 1 to 3

	emp_id [PK] integer	emp_name character varying (100)	salary integer
1	101	Amit	25000
2	103	Rahul	45000
3	105	Arjun	50000

Successfully run. Tota

Learning Outcome

After completing this experiment, the learner will be able to:

- Understand conditional logic in SQL queries.
- Use the MOD (%) operator for numerical analysis.
- Classify numeric values into odd and even categories.
- Write efficient SQL queries for payroll data analysis.
- Apply SQL logic in real-world enterprise environments.